LPJ-EOSIM

Model ecosystem

lpj-eosim.github.io

MIP feedback & yearly cycles: revise & rerun

Global regression test vs KGO: revise & re-push

Local debug: profile & develop

GitHub

Local debug

- *1–5 cells*
- *Targeted debugging on a range of biomes*
- *Enables fast research turnaround & diagnostics*

Local run

Model · C

LPJ-EOSIM

- *Carbon, N, wetland CH_4, fire*
- *Embarrassingly parallel*
- *Merged PRs: comprehensive benchmarking + good science*
- *Releases for data pushes + MIPs*

snapshot + compile

Driver data

`lpj_download`

- *Downloads & formats reanalysis for LPJ*
- *Range & gap tests on climate data*

Forcing

NetCDF

Python API

`pipeline`

- *Driver registry*
- *Compiles + launches runs*
- *Injects standardized configuration into runs*

drives

Release

Ingest

DOI **Zenodo archive**

- *Push LPJ drivers to Zenodo for reproducibility*

Python package

`lpj-eosim-run-scripts`

- *Drives MIP pushes*
- *Reproducible analysis and run environments for LPJ*

Claude Code skill

`run-lpj`

- *Agent recipe: run + debug LPJ*
- *Runs all three repos above*

Open PR

Launch

SLURM / AWS

lpj-ci · GitHub Actions

Global regression test

- *PR vs known-good (KGO)*
- *Bitwise identity check*
- *Diffs need an explanation*
- *Merged runs promoted to new KGO*

SLURM jobs

Production runs

- *Global grid, 0.0083° to 0.5°*
- *Automatic ILAMB benchmarking*
- *Regression plots*

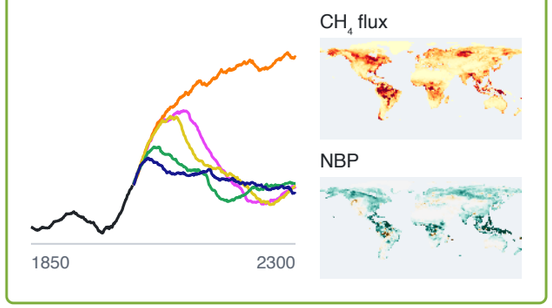
Publish on success

`lpj-registry`

Wasabi S3 · object lock

- *Git provenance: LPJ + pipeline*
- *Frozen artifacts, campaign tags*
- *Web UI: compare vs every past run*

lpj-registry · web UI



Submit

per protocol

Submissions & products

TRENDY
Global Carbon Budget, yearly

NRT carbon budget
Low-latency carbon budget

Global Methane Budget
Global Carbon Project, wetland CH_4

WIEMIP
CMIP Community MIP

NMIP
Global N_2O budget

NASA LP DAAC
Low-latency wetland CH_4 + NBP
Cloud-Optimized GeoTIFF (COG)
Daily pushes, 4-month latency

S2S forecasting
Subseasonal to seasonal forecasting of carbon fluxes